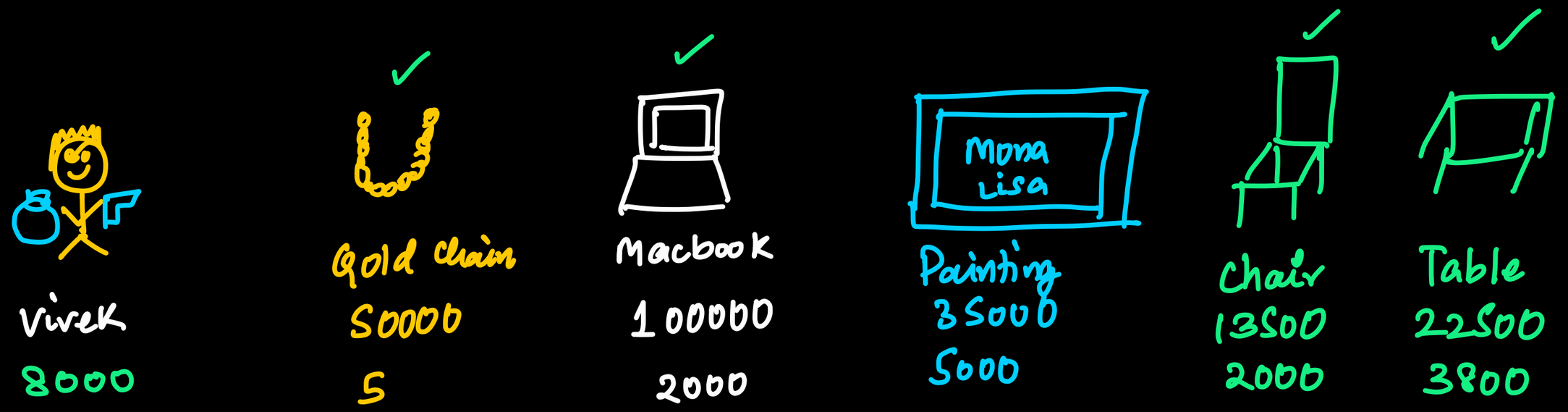
Problem

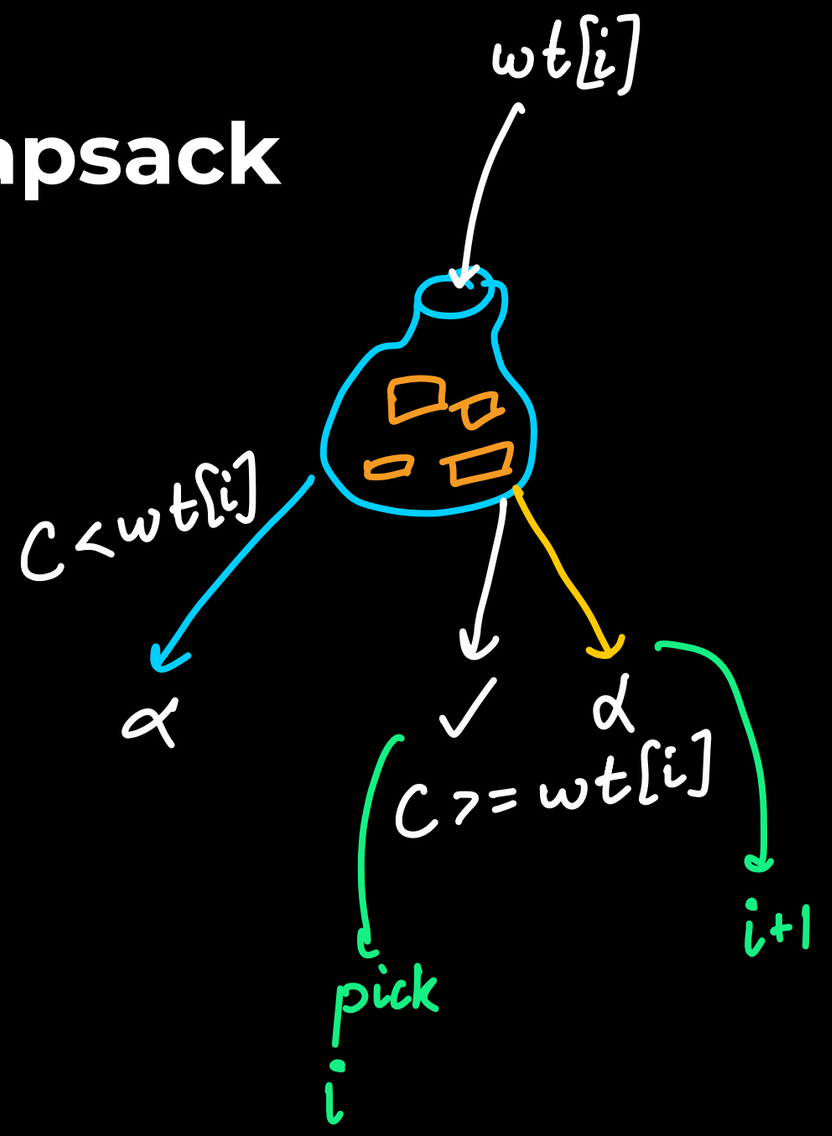
Unbounded Knapsack

unlimited supply of each item



Problem

Unbounded Knapsack



Problem

Unbounded Knapsack

Capacity	Item 1	Item 2
	100	76
8	5	4
	Skip it	choose 2 of these

Problem

Coin Change

Problem

Target Sum

Problem

Longest Common Subsequence (LCS)

s1 = a b c d e f g m

s2 = r b d m e o e z y g

b d e g
b d e g] len = 4

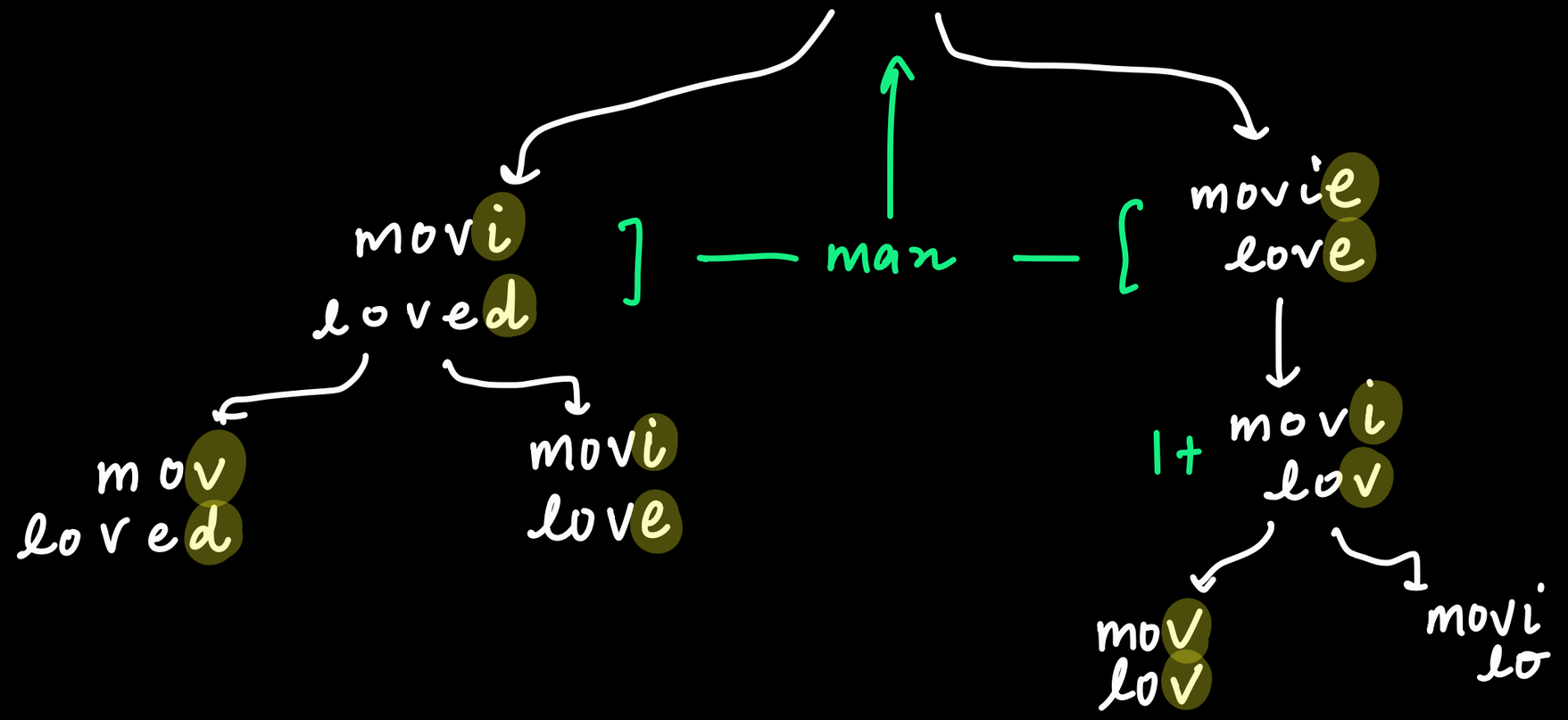
Problem

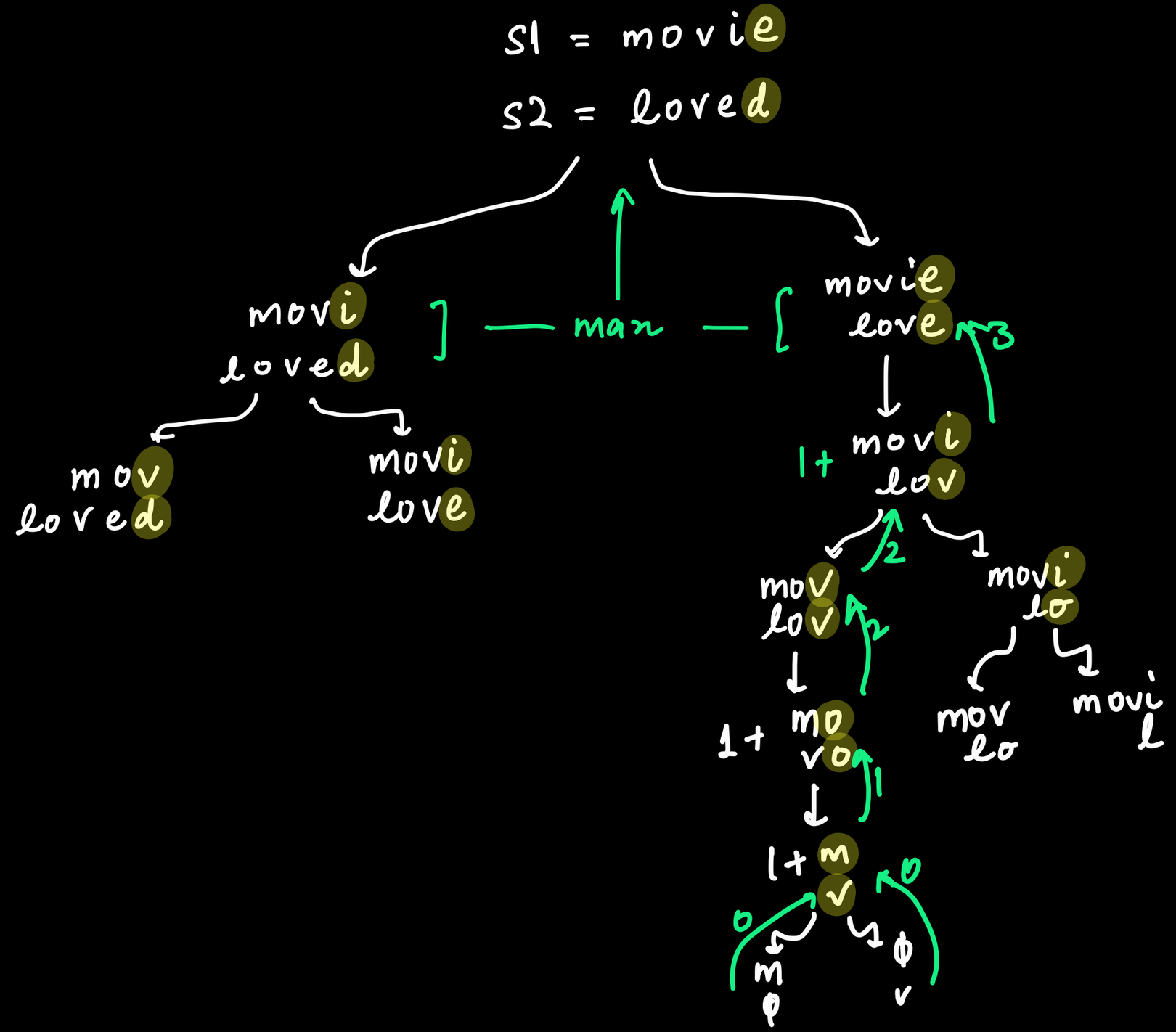
ove - 3

Longest Common Subsequence

s1 = movie

s2 = loved




s1 = movie
s2 = loved
movi
loved
mov
loved
movi
love
max
movie
love
3
1+ movi
lov
2
mov
lov
2
movi
lo
1+ mo
vo
1
mov
lo
movi
l
1+ m
v
0
0
m
φ
φ
v

Problem

Longest Palindromic Subsequence

LCS

s = a b c d e b g a

rev = a g b e d c b a

len = s

LPS(s) = LCS(s, rev(s));

Problem

Minimum insertions to make string Palindrome

S = "s a g a r"

r s

2 insertions

s r a g a r s

return S.length - LPS(S);

Homework

Minimum deletions to make string Palindrome